
Quantitative Seismology Assignment 1

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Assignment: Number 1

Problem 1 Kronecker Delta and Levi-Civita Symbol Identities

(1) $\delta_{ii} = 3$

$$\delta_{ii} = \delta_{11} + \delta_{22} + \delta_{33} = 1 + 1 + 1 = 3$$

(2) $\delta_{ij}\varepsilon_{ijk} = 0$

$$\delta_{ij}\varepsilon_{ijk} = \varepsilon_{iik} = 0$$

(3) $\varepsilon_{ipq}\varepsilon_{j pq} = 2\delta_{ij}$

Using the determinant property:

$$\varepsilon_{ipq}\varepsilon_{j pq} = \det \begin{pmatrix} \mathbf{e}_i^\top \\ \mathbf{e}_p^\top \\ \mathbf{e}_q^\top \end{pmatrix} \cdot \det (\mathbf{e}_j \quad \mathbf{e}_p \quad \mathbf{e}_q) = \det \left(\begin{pmatrix} \mathbf{e}_i^\top \\ \mathbf{e}_p^\top \\ \mathbf{e}_q^\top \end{pmatrix} \cdot (\mathbf{e}_j \quad \mathbf{e}_p \quad \mathbf{e}_q) \right) = \det \begin{pmatrix} \delta_{ij} & \delta_{ip} & \delta_{iq} \\ \delta_{pj} & \delta_{pp} & \delta_{pq} \\ \delta_{qj} & \delta_{qp} & \delta_{qq} \end{pmatrix}$$

Expanding the determinant:

$$\begin{aligned} &= \delta_{ij}(\delta_{pp}\delta_{qq} - \delta_{pq}\delta_{pq}) - \delta_{ip}(\delta_{pj}\delta_{qq} - \delta_{pq}\delta_{qj}) + \delta_{iq}(\delta_{pj}\delta_{qp} - \delta_{qj}\delta_{pp}) \\ &= 3 * 3\delta_{ij} - \delta_{ij}\delta_{pq}\delta_{pq} - \delta_{ij}\delta_{qq} + \delta_{ip}\delta_{pj} + \delta_{iq}\delta_{jq} - \delta_{ij}\delta_{pp} \\ &= 9\delta_{ij} - \delta_{pp}\delta_{ij} - \delta_{qq}\delta_{ij} + \delta_{ij} + \delta_{ij} - \delta_{pp}\delta_{ij} \\ &= (9 - 3 - 3 + 1 + 1 - 3)\delta_{ij} \\ &= 2\delta_{ij} \end{aligned}$$

(4) $\varepsilon_{ipq}\varepsilon_{ipq} = 6$

$$\varepsilon_{ipq}\varepsilon_{ipq} = \det \begin{pmatrix} \mathbf{e}_i^\top \\ \mathbf{e}_p^\top \\ \mathbf{e}_q^\top \end{pmatrix} \cdot \det (\mathbf{e}_i^\top \quad \mathbf{e}_p^\top \quad \mathbf{e}_q^\top) = \det \left(\begin{pmatrix} \mathbf{e}_i^\top \\ \mathbf{e}_p^\top \\ \mathbf{e}_q^\top \end{pmatrix} \cdot (\mathbf{e}_i \quad \mathbf{e}_p \quad \mathbf{e}_q) \right) = \det \begin{pmatrix} \delta_{ii} & \delta_{ip} & \delta_{iq} \\ \delta_{pi} & \delta_{pp} & \delta_{pq} \\ \delta_{qi} & \delta_{qp} & \delta_{qq} \end{pmatrix}$$

Expanding:

$$\begin{aligned} &= \delta_{ii}(\delta_{pp}\delta_{qq} - \delta_{pq}\delta_{qp}) - \delta_{ip}(\delta_{pi}\delta_{qq} - \delta_{pq}\delta_{qi}) + \delta_{iq}(\delta_{pi}\delta_{qp} - \delta_{pp}\delta_{qi}) \\ &= 3(3 \times 3 - \delta_{pp}) - \delta_{ip}\delta_{pi}\delta_{qq} + \delta_{ip}\delta_{pq}\delta_{qi} + \delta_{iq}\delta_{pi}\delta_{qp} - \delta_{iq}\delta_{pp}\delta_{qi} \\ &= 3(3 \times 3 - 3) - \delta_{pp}\delta_{qq} + \delta_{iq}\delta_{qi} + \delta_{pq}\delta_{qp} - \delta_{qq}\delta_{pp} \\ &= 18 - 3 \times 3 + 3 + 3 - 3 \times 3 \\ &= 6 \end{aligned}$$

Problem 2 Tensor Proofs

(1) Prove $\mathbf{a} \cdot \mathfrak{S} \cdot \mathbf{b} = -\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{a}$, where $\mathfrak{S} = \varepsilon_{ijk} \mathbf{e}_i \mathbf{e}_j \mathbf{e}_k$ is a third-order tensor.

$$\begin{aligned}
 \mathbf{a} \cdot \mathfrak{S} \cdot \mathbf{b} &= a_m \mathbf{e}_m \cdot \varepsilon_{ijk} \mathbf{e}_i \mathbf{e}_j \mathbf{e}_k \cdot b_n \mathbf{e}_n \\
 &= \varepsilon_{ijk} a_m \delta_{mi} b_n \delta_{kn} \mathbf{e}_j \\
 &= \varepsilon_{ijk} a_i b_k \mathbf{e}_j \\
 &= \varepsilon_{kij} b_k a_i \mathbf{e}_j \\
 &= \mathbf{b} \times \mathbf{a} \\
 &= -\varepsilon_{ikj} a_i b_k \mathbf{e}_j \\
 &= -\mathbf{a} \times \mathbf{b}
 \end{aligned}$$

(2) Prove $\mathbf{a} \cdot \mathbf{a}$ is invariant under coordinate rotation

$$\begin{aligned}
 \mathbf{a} \cdot \mathbf{a} &= a_i \mathbf{e}_i \cdot a_j \mathbf{e}_j \\
 &= (\alpha_{ki} a'_k) (\alpha_{mi} \mathbf{e}'_m) \cdot (\alpha_{pj} a'_p) (\alpha_{nj} \mathbf{e}'_n) \\
 &= (\alpha_{ki} \alpha_{mi}) a'_k a'_p (\alpha_{pj} \alpha_{nj}) (\mathbf{e}'_m \cdot \mathbf{e}'_n) \\
 &= \delta_{km} a'_k a'_p \delta_{pn} \delta_{mn} \\
 &= a'_m a'_n \delta_{mn} \\
 &= a'_m a'_m = \mathbf{a}' \cdot \mathbf{a}'
 \end{aligned}$$

Problem 3 Anti-symmetric Tensor and Dual Vector

Given $A_k = \frac{1}{2} \varepsilon_{ijk} B_{ij}$ where $B_{ij} = -B_{ji}$. Prove: $B_{mn} = \varepsilon_{mnk} A_k$.

$$\begin{aligned}
 \varepsilon_{mnk} A_k &= \frac{1}{2} \varepsilon_{mnk} \varepsilon_{ijk} B_{ij} \\
 &= \frac{1}{2} (\delta_{mi} \delta_{nj} - \delta_{mj} \delta_{ni}) B_{ij} \\
 \varepsilon_{mnk} A_k &= \frac{1}{2} (B_{mn} - B_{nm}) \\
 &= B_{mn} \quad (\because B_{nm} = -B_{mn})
 \end{aligned}$$

Problem 4 Symmetric Tensor Condition

Prove: $\varepsilon_{kij} C_{ij} = 0$ if and only if $C_{ij} = C_{ji}$.

Necessity: From $\varepsilon_{kij} C_{ij} = 0$, multiply by ε_{mnk} | **Sufficiency:** If $C_{ij} = C_{ji}$, then

$$\begin{aligned}
 \varepsilon_{mnk} \varepsilon_{ijk} C_{ij} &= 0 & \varepsilon_{kij} C_{ij} &= -\varepsilon_{kji} C_{ij} = -\varepsilon_{kji} C_{ji} = -\varepsilon_{kij} C_{ij} \\
 (\delta_{mi} \delta_{nj} - \delta_{mj} \delta_{ni}) C_{ij} &= 0 & &\Rightarrow 2\varepsilon_{kij} C_{ij} = 0 \\
 C_{mn} - C_{nm} &= 0 & &\Rightarrow \varepsilon_{kij} C_{ij} = 0 \\
 \Rightarrow C_{mn} &= C_{nm}
 \end{aligned}$$

Problem 5 Vector Identities via ε - δ Identity

(1) $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c})$

$$\begin{aligned} (\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) &= (\varepsilon_{ijk} a_i b_j \mathbf{e}_k) \cdot (\varepsilon_{mnp} c_m d_n \mathbf{e}_p) \\ &= \varepsilon_{ijk} \varepsilon_{mnp} a_i b_j c_m d_n \delta_{kp} = \varepsilon_{ijk} \varepsilon_{mnk} a_i b_j c_m d_n \\ &= (\delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}) a_i b_j c_m d_n = a_m c_m b_n d_n - a_n d_n b_m c_m \\ &= (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c}) \end{aligned}$$

(2) $\nabla \times \nabla \psi = 0$

$$\begin{aligned} \nabla \times \nabla \psi &= \nabla \times (\psi_{,i} \mathbf{e}_i) = \varepsilon_{ijp} \psi_{,ji} \mathbf{e}_p \\ &= \frac{1}{2} (\varepsilon_{ijp} \psi_{,ji} \mathbf{e}_p + \varepsilon_{ijp} \psi_{,ji} \mathbf{e}_p) = \frac{1}{2} (-\varepsilon_{jip} \psi_{,ji} \mathbf{e}_p + \varepsilon_{ijp} \psi_{,ij} \mathbf{e}_p) \\ &= \frac{1}{2} (-\varepsilon_{mnp} \psi_{,mn} \mathbf{e}_p + \varepsilon_{mnp} \psi_{,mn} \mathbf{e}_p) = 0 \end{aligned}$$

(3) $\nabla \cdot (\nabla \times \mathbf{a}) = 0$

$$\begin{aligned} \nabla \cdot (\nabla \times \mathbf{a}) &= \nabla \cdot (\varepsilon_{ijk} a_{j,i} \mathbf{e}_k) \\ &= \mathbf{e}_p \cdot \varepsilon_{ijk} a_{j,i} \mathbf{e}_k = \varepsilon_{ijk} a_{j,i} \delta_{kp} = \varepsilon_{ijp} a_{j,i} \\ &= \frac{1}{2} (\varepsilon_{ijp} a_{j,i} + \varepsilon_{ijp} a_{j,i}) \\ &= \frac{1}{2} (\varepsilon_{ijp} a_{j,i} - \varepsilon_{pji} a_{j,pi}) \\ &= \frac{1}{2} (\varepsilon_{mnp} a_{n,mp} - \varepsilon_{mnp} a_{n,mp}) \\ &= 0 \end{aligned}$$

(4) $\nabla \times (\nabla \times \mathbf{a}) = \nabla (\nabla \cdot \mathbf{a}) - \nabla^2 \mathbf{a}$

$$\begin{aligned} \nabla \times (\nabla \times \mathbf{a}) &= \nabla \times (\varepsilon_{ijk} a_{j,i} \mathbf{e}_k) \\ &= \varepsilon_{mnp} \varepsilon_{ijn} a_{j,im} \mathbf{e}_p \\ &= \varepsilon_{pmn} \varepsilon_{ijn} a_{j,im} \mathbf{e}_p = (\delta_{pi} \delta_{mj} - \delta_{pj} \delta_{mi}) a_{j,im} \mathbf{e}_p \\ &= \delta_{pi} \delta_{mj} a_{j,im} \mathbf{e}_p - \delta_{pj} \delta_{mi} a_{j,im} \mathbf{e}_p \\ &= a_{m,pm} \mathbf{e}_p - a_{p,mm} \mathbf{e}_p = a_{m,mp} \mathbf{e}_p - a_{p,mm} \mathbf{e}_p \\ &= \nabla (\nabla \cdot \mathbf{a}) - \nabla^2 \mathbf{a} \end{aligned}$$

(5) $\nabla \times (\psi \mathbf{a}) = \nabla \psi \times \mathbf{a} + \psi (\nabla \times \mathbf{a})$

$$\begin{aligned} \nabla \times (\psi \mathbf{a}) &= \varepsilon_{ijk} \frac{\partial}{\partial x_i} (\psi a_j) \mathbf{e}_k \\ &= \varepsilon_{ijk} (\psi_{,i} a_j + \psi a_{j,i}) \mathbf{e}_k \\ &= \varepsilon_{ijk} \psi_{,i} a_j \mathbf{e}_k + \psi \varepsilon_{ijk} a_{j,i} \mathbf{e}_k \\ &= \nabla \psi \times \mathbf{a} + \psi (\nabla \times \mathbf{a}) \end{aligned}$$

$$(6) \nabla(\mathbf{a} \cdot \mathbf{b}) = (\mathbf{a} \cdot \nabla)\mathbf{b} + (\mathbf{b} \cdot \nabla)\mathbf{a} + \mathbf{a} \times (\nabla \times \mathbf{b}) + \mathbf{b} \times (\nabla \times \mathbf{a})$$

$$\begin{aligned} \nabla(\mathbf{a} \cdot \mathbf{b}) &= \frac{\partial}{\partial x_i} (a_j b_j) \mathbf{e}_i \\ &= a_{j,i} b_j \mathbf{e}_i + a_j b_{j,i} \mathbf{e}_i \\ &= a_{j,i} b_j \mathbf{e}_i - a_{i,j} b_j \mathbf{e}_i + a_{i,j} b_j \mathbf{e}_i + a_j b_{j,i} \mathbf{e}_i - a_j b_{i,j} \mathbf{e}_i + a_j b_{i,j} \mathbf{e}_i \\ &= (\delta_{ip} \delta_{jq} - \delta_{iq} \delta_{jp}) a_{i,j} b_p \mathbf{e}_q + b_j a_{i,j} \mathbf{e}_i + (\delta_{ij} \delta_{pq} - \delta_{ip} \delta_{jq}) a_i b_{j,p} \mathbf{e}_q + a_j b_{i,j} \mathbf{e}_i \\ &= \varepsilon_{ijk} \varepsilon_{pqk} a_{i,j} b_p \mathbf{e}_q + b_j \frac{\partial}{\partial x_j} a_i \mathbf{e}_i + \varepsilon_{iqk} \varepsilon_{jpk} a_i b_{j,p} \mathbf{e}_q + a_j \frac{\partial}{\partial x_j} b_i \mathbf{e}_i \\ &= \varepsilon_{pkq} b_p (\varepsilon_{jik} a_{i,j}) \mathbf{e}_q + (\mathbf{b} \cdot \nabla) \mathbf{a} + \varepsilon_{ikq} a_i (\varepsilon_{pj k} b_{j,p}) \mathbf{e}_q + (\mathbf{a} \cdot \nabla) \mathbf{b} \\ &= \mathbf{b} \times (\nabla \times \mathbf{a}) + (\mathbf{b} \cdot \nabla) \mathbf{a} + \mathbf{a} \times (\nabla \times \mathbf{b}) + (\mathbf{a} \cdot \nabla) \mathbf{b} \end{aligned}$$

$$(7) \nabla \times (\mathbf{a} \times \mathbf{b}) = \mathbf{a}(\nabla \cdot \mathbf{b}) - \mathbf{b}(\nabla \cdot \mathbf{a}) + (\mathbf{b} \cdot \nabla) \mathbf{a} - (\mathbf{a} \cdot \nabla) \mathbf{b}$$

$$\begin{aligned} \nabla \times (\mathbf{a} \times \mathbf{b}) &= \varepsilon_{ijk} \frac{\partial}{\partial x_i} (\varepsilon_{mnj} a_m b_n) \mathbf{e}_k \\ &= \varepsilon_{kij} \varepsilon_{mnj} (a_{m,i} b_n + a_m b_{n,i}) \mathbf{e}_k \\ &= (\delta_{km} \delta_{in} - \delta_{kn} \delta_{im}) (a_{m,i} b_n + a_m b_{n,i}) \mathbf{e}_k \\ &= a_{m,n} b_n \mathbf{e}_m - a_{m,m} b_n \mathbf{e}_n + a_m b_{n,n} \mathbf{e}_m - a_m b_{n,m} \mathbf{e}_n \\ &= \left(b_n \frac{\partial}{\partial x_n} \right) a_m \mathbf{e}_m - b_n \mathbf{e}_n \frac{\partial}{\partial x_m} (a_m) \\ &\quad + a_m \mathbf{e}_m \left(\frac{\partial}{\partial x_n} b_n \right) - \left(a_m \frac{\partial}{\partial x_m} \right) b_n \mathbf{e}_n \\ &= (\mathbf{b} \cdot \nabla) \mathbf{a} - \mathbf{b}(\nabla \cdot \mathbf{a}) + \mathbf{a}(\nabla \cdot \mathbf{b}) - (\mathbf{a} \cdot \nabla) \mathbf{b} \end{aligned}$$

Problem 6 Simplification of Elastodynamic Equation for Isotropic medium

General elastodynamic equation:

$$\rho \ddot{u}_i = \frac{\partial}{\partial x_j} [c_{ijkl} u_{k,l}] + f_i, \quad c_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$$

$$\begin{aligned} \frac{\partial}{\partial x_j} [c_{ijkl} u_{k,l}] &= c_{ijkl} u_{k,lj} = [\lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})] u_{k,lj} \\ &= \lambda u_{k,ki} + \mu u_{i,ll} + \mu u_{k,ik} = (\lambda + 2\mu) u_{k,ki} - \mu (u_{k,ik} - u_{i,kk}) \\ &= (\lambda + 2\mu) (\nabla (\nabla \cdot \mathbf{u}))_i - \mu (\delta_{il} \delta_{jk} - \delta_{ik} \delta_{jl}) u_{k,lj} \\ &= (\lambda + 2\mu) (\nabla (\nabla \cdot \mathbf{u}))_i - \mu (\varepsilon_{ijm} \varepsilon_{lkm}) u_{k,lj} \\ &= (\lambda + 2\mu) (\nabla (\nabla \cdot \mathbf{u}))_i - \mu \left[\varepsilon_{ijm} \frac{\partial}{\partial x_j} \left(\varepsilon_{lkm} \frac{\partial u_k}{\partial x_l} \right) \right] \\ &= (\lambda + 2\mu) (\nabla (\nabla \cdot \mathbf{u}))_i - \mu \left[\varepsilon_{jmi} \frac{\partial}{\partial x_j} \left(\varepsilon_{lkm} \frac{\partial u_k}{\partial x_l} \right) \right] \\ &= (\lambda + 2\mu) \nabla (\nabla \cdot \mathbf{u}) - \mu [\nabla \times (\nabla \times \mathbf{u})]_i \end{aligned}$$

Therefore:

$$\rho \ddot{\mathbf{u}} = (\lambda + 2\mu) \nabla (\nabla \cdot \mathbf{u}) - \mu \nabla \times (\nabla \times \mathbf{u}) + \mathbf{f}$$

Problem 7 Inversion of Stress-Strain Relation

Given: $\tau_{ij} = \lambda e_{kk} \delta_{ij} + 2\mu e_{ij}$. Derive: $e_{ij} = -\frac{\lambda}{2\mu(3\lambda + 2\mu)} \tau_{kk} \delta_{ij} + \frac{1}{2\mu} \tau_{ij}$

For $i = j$, multiply by δ_{ij} (take the trace):

$$\begin{aligned}\tau_{ij} \delta_{ij} &= \lambda e_{kk} \delta_{ij} \delta_{ij} + 2\mu e_{ij} \delta_{ij} \\ \tau_{ii} &= 3\lambda e_{kk} + 2\mu e_{ii} \\ \tau_{ii} &= (3\lambda + 2\mu) e_{kk} \\ \Rightarrow e_{kk} &= \frac{\tau_{ii}}{3\lambda + 2\mu} = \frac{\tau_{kk}}{3\lambda + 2\mu}\end{aligned}$$

Substitute back into the original equation:

$$\begin{aligned}\tau_{ij} &= \lambda e_{kk} \delta_{ij} + 2\mu e_{ij} \\ 2\mu e_{ij} &= \tau_{ij} - \lambda \left(\frac{\tau_{kk}}{3\lambda + 2\mu} \right) \delta_{ij} \\ e_{ij} &= \frac{\tau_{ij}}{2\mu} - \frac{\lambda}{2\mu(3\lambda + 2\mu)} \tau_{kk} \delta_{ij}\end{aligned}$$

Problem 8 Derivation of Traction Vector Formula

Given: $\mathbf{T}(\mathbf{u}, \mathbf{n}) = \mathbf{n} \cdot \boldsymbol{\tau}$, where $\tau_{ij} = \lambda (\nabla \cdot \mathbf{u}) \delta_{ij} + \mu (u_{i,j} + u_{j,i})$

Derive: $\mathbf{T}(\mathbf{u}, \mathbf{n}) = \lambda (\nabla \cdot \mathbf{u}) \mathbf{n} + \mu \left(2 \frac{\partial \mathbf{u}}{\partial n} + \mathbf{n} \times (\nabla \times \mathbf{u}) \right)$

$$\begin{aligned}T_j &= n_k \tau_{kj} \\ &= n_k [\lambda (\nabla \cdot \mathbf{u}) \delta_{kj} + \mu (u_{k,j} + u_{j,k})] \\ &= n_k \lambda (\nabla \cdot \mathbf{u}) \delta_{kj} + \mu n_k (u_{k,j} + u_{j,k}) \\ &= \lambda (\nabla \cdot \mathbf{u}) n_j + \mu n_k u_{k,j} + \mu n_k u_{j,k} \\ &= \lambda (\nabla \cdot \mathbf{u}) n_j + \mu (n_k u_{k,j} + 2n_k u_{j,k} - n_k u_{j,k}) \\ &= \lambda (\nabla \cdot \mathbf{u}) n_j + \mu \left[2n_k \frac{\partial u_j}{\partial x_k} + n_k (u_{k,j} - u_{j,k}) \right] \\ &= \lambda (\nabla \cdot \mathbf{u}) n_j + \mu [2(\mathbf{n} \cdot \nabla) u_j + n_k (u_{k,j} - u_{j,k})] \\ &= \lambda (\nabla \cdot \mathbf{u}) n_j + \mu \left[2(\mathbf{n} \cdot \nabla) u_j + [(\delta_{ik} \delta_{pq} - \delta_{ip} \delta_{kq}) n_i u_{k,p} e_q]_j \right] \\ &= \lambda (\nabla \cdot \mathbf{u}) n_j + \mu \left[2(\mathbf{n} \cdot \nabla) u_j + [\varepsilon_{iqm} \varepsilon_{kpm} n_i u_{k,p} e_q]_j \right] \\ &= \lambda (\nabla \cdot \mathbf{u}) n_j + \mu \left[2(\mathbf{n} \cdot \nabla) u_j + [\varepsilon_{imq} n_i \varepsilon_{pkm} u_{k,p} e_q]_j \right] \\ &= \lambda (\nabla \cdot \mathbf{u}) \mathbf{n} + \mu [2(\mathbf{n} \cdot \nabla) \mathbf{u} + \mathbf{n} \times (\nabla \times \mathbf{u})]\end{aligned}$$

where $\frac{\partial \mathbf{u}}{\partial n} = (\mathbf{n} \cdot \nabla) \mathbf{u}$